Using NBA Playstyles to Predict Wins

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1 Abstract

This paper investigates whether NBA team playstyles, derived through unsupervised clustering, carry substantial predictive power over game outcomes beyond conventional team quality metrics. Using a rolling 10-game window of stylistic statistics drawn from 30 years of NBA regular season data (1996–2026), we apply K-Means clustering to identify five interpretable playstyle archetypes. We then encode stylistic matchups as an explicit interaction feature within an XGBoost classifier, testing the hypothesis that certain playstyles hold structural advantages over others in a "rock-paper-scissors" fashion. Our model achieves the highest ROC-AUC (0.6837) of all evaluated approaches, surpassing both a win-percentage baseline and an offense-only logistic regression, suggesting that playstyle interaction contributes genuine probabilistic signal. However, playstyle does not improve raw predictive accuracy, and SHAP analysis confirms that rolling win percentage remains the overwhelmingly dominant predictor. We conclude that while stylistic matchups offer marginal incremental value, team quality is the primary driver of NBA game outcomes.

2 Introduction

The National Basketball Association (NBA) has a history which reaches back decades. Throughout it, teams have adopted and invented various "playstyles". Some teams are slow, methodical, and defense-oriented while others want to get the ball and shoot as quickly as possible. Both of these playstyles have had their place in the NBA, but each playstyle comes with advantages and disadvantages.

This begs the question: when certain playstyles match up, who holds the advantage? Do great defensive teams stifle great offenses? Do fast paced teams leave slower paced teams in the dust? If two teams have the same winning percentage, could one team's playstyle give them the upper-hand in a head-to-head matchup? Is there a "rock-paper-scissors" like effect in the NBA with different playstyles? That is to say, is it possible that one playstyle "A" is more advantageous against playstyle "B", while being disadvantageous against playstyle "C"?

This method of aggregating playstyles differs from typical approaches to predicting win outcomes in the NBA. While most models evaluate a team based on dozens of individual stats, this clustering technique puts them into context. It allows us to group teams and evaluate their interactions more holistically and detect trends among the different playstyles. Analyzing the interactions between these playstyles could provide NBA teams a better idea of the most optimal playstyles to pursue going forward.

2.1 Hypotheses

We hypothesize that a clustering algorithm such as "K-Means" can create groups of teams which align with real, popular playstyles from recent NBA history (1996–2026). These clusters will reflect real viable strategies which teams have embodied over the years. Additionally, we hypothesize that these playstyles will hold more predictive power than simpler baseline models.

3 Dataset

Our dataset was acquired from user "Eoin A Moore" on Kaggle.com under the file name `TeamStatisticsExtended.csv`. This dataset aggregates official boxscores from all NBA games between 1996–2026 and lists team statistics ranging from basic statistics (points, assists, rebounds, etc.) to more advanced statistics (assist percentage, pace, true shooting, etc.). The dataset has a total of 67,430 instances across 30 years with 100 columns covering both team information and statistics.

4 Preprocessing

4.1 Game Selection

Before we were able to perform our playstyle clustering, we needed to do some filtering and make design decisions. First, we opted to focus only on "regular season" games from our dataset. For each NBA season in our dataset, every team played 82 "regular season" games, but afterwards the 16 best performing teams are invited to the "playoffs" where they play anywhere from 4 to 28 games. We quickly decided that including this data would introduce noise into our model because it would make it so some teams were overrepresented. We also decided that the sample size of 4–28 possible games for 16 teams would not be enough to perform any conclusive analysis on separately. For these reasons, we strictly focused on the "regular season."

4.2 Rolling Averages

Our next design decision was whether to consider cumulative season statistics or a rolling average. The issue with cumulative season statistics is the problem of data leakage. If we are asked to predict the 50th game of the season for a team based on their full season of data, that 50th game has already been used to train the model. And using last season's data to predict the next season would misrepresent teams who suffered injuries, performed trades, or had playstyle changes.

With rolling averages, we were able to avoid this problem of data leakage. For each game we want to predict, that team's playstyle is determined by their behavior in their last X games, X being our rolling averages window. We settled on a window of 10 games, because this would help capture any recent big changes to a team's playstyle that could be due to injury or trades. Since our games are predicted based on the previous behavior in the team's X prior games that season, we needed to drop the first X games of each team's season, as they hadn't played enough to effectively predict upon. A larger window size could have produced less noise in our evaluations, but it would be less flexible to drastic changes in playstyle and would cut into the available data for our model. In making this decision, our model will be able to make predictions on games 11–82 of an NBA team's season.

5 Playstyle Clustering

5.1 K-means Algorithm

To define playstyles, we used K-Means Clustering; an unsupervised algorithm that splits observations into k groups by minimizing the within-cluster sum of squared distances from each data point to its cluster centroid. Each game data point, represented by a team's 10-game rolling averages going into that game, gets assigned to the nearest centroid, and centroids are iteratively updated until convergence.

K-Means was chosen for a couple of reasons. First, its hard cluster assignments produce clean, interpretable labels. Each team in each game belongs to exactly one playstyle, which was necessary when we had to put playstyles head-to-head in our EDA and modeling. Second, K-Means is well-suited to our NBA statistics since the rolling averages are all continuous, numeric, and scalable, which is a requirement of the algorithm.

5.2 Feature Selection

Selecting suitable features for our clustering considered much thought and domain knowledge. Although `offensiveRating` and `defensiveRating` are well-known metrics in basketball analytics, they were deliberately excluded. Both are composite summary statistics that collapse the entire outcome of possessions into a single number, already partially reflecting all the other features in our set. Having an offensive rating higher than your defensive rating, implies that that team has an average margin of victory greater than 0 and vice versa. Including these stats would cause the clustering to organize around overall team quality rather than stylistic identity. By excluding these ratings and instead using the underlying shot location, pace, and possession metrics, we force the clusters to capture *how* a team plays rather than *how well* it plays.

We ultimately decided to use the features listed in Table 1. Besides pace, all of these statistics are proportions and do not-

Table 1: Style Features Used for Playstyle Clustering

Feature	Description
`pace`	Estimated number of possessions per 48 minutes; measures how fast a team plays
`assistPercentage`	Percentage of field goals that were assisted; reflects ball movement and offensive system
`percentFieldGoalAttempts2Point`	Share of all field goal attempts that were 2-point shots; captures shot selection philosophy
`percentFieldGoalAttempts3Point`	Share of all field goal attempts that were 3-point shots; key indicator of modern vs. traditional offense
`percentPoints2Point`	Proportion of total points scored from 2-point field goals
`percentPoints2PointMidRange`	Proportion of total points scored from mid-range 2-point shots; hallmark of the Grit & Grind era
`percentPoints3Point`	Proportion of total points scored from 3-point field goals
`percentPointsFastBreak`	Proportion of total points scored in transition; reflects pace and defensive rebounding aggressiveness
`percentPointsFreeThrow`	Proportion of total points scored from the free throw line; captures foul-drawing tendencies
`percentPointsOffTurnovers`	Proportion of points scored directly off opponent turnovers; reflects defensive pressure and transition offense
`percentPointsInPaint`	Proportion of total points scored in the paint (16ft $\times$ 19ft rectangle around the basket)
`trueShootingPercentage`	Overall shooting efficiency accounting for point values of 2-pointers, 3-pointers, and free throws
`teamTurnoverPercentage`	Percentage of a team's own possessions that end in a turnover; measures ball security
`opponentTurnoverPercentage`	Percentage of opponent possessions forced into a turnover; measures defensive pressure
`opponentEffectiveFieldGoal-Percentage`	Opponent's effective field goal percentage (adjusted for 3-pointers); measures quality of a team's defense against shooting

-give any indication of how well a team performs. Rather, the features simply reflect tendencies, strengths, and weaknesses which come with a team's playstyle.

5.3 Creating Clusters

To justify our choice of k, we used the elbow method. We ran K-Means across $k = 2$ through $k = 8$ and plotted the Sum of Squared Errors (the total within-cluster variance) at each step. The principle is to identify the point where adding more clusters stops producing meaningful reductions in SSE, which visually appears as a "bend" or "elbow" in the curve.

The elbow in our curve appears at $k = 5$. At this point, SSE drops substantially from $k = 4$ to $k = 5$, but the curve flattens into a near-linear decay from $k = 6$ onwards, indicating that additional clusters are merely splitting existing groups rather than discovering genuinely new structure. Beyond the statistical justification, $k = 5$ aligns naturally with five interpretable basketball playstyles. For our purposes, this is an important criteria since cluster labels need to convey existing and observable basketball playstyles.

After running K-Means with $k = 5$, the resulting clusters, in order from most frequent to least, are: "Slow & Balanced", "Rim Runners", "Grit & Grind", "Run & Gun" and "Quick & Balanced." Descriptions of these playstyles are available in Table 2. Each cluster was well-represented with between 15,332–9,429 instances per cluster.

6 Exploratory Data Analysis

6.1 Playstyle Changes Over Time

In our EDA, we quickly found that playstyle prevalence was heavily influenced by the year. As seen in Figure 1, the distribution of playstyles has drastically shifted over the past 30 years of the NBA. Fast-paced and heavy 3-point shooting teams have taken over from slower, mid-range focused teams. Since 2020, teams have almost entirely played in these fast-paced, high 3-pointer playstyles, but before 2005 they were nearly non-existent. This lines up with the common sentiment around the NBA; teams learned that close in shots and 3-pointers are the most effective by far and have curated their offense to exclude inefficient mid-range shooting.

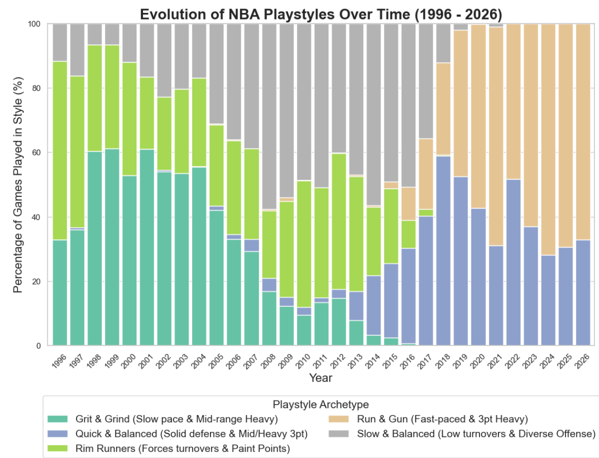

Figure 1: Playstyle distribution over time

"Grit & Grind" teams earned around 26% of their points from mid-range, while modern "Run & Gun" teams typically got less than 8%. Additionally, the post-2020 playstyles averaged nearly 6 more possessions per game than all of the pre-2005 playstyles.

This difference appeared in our PCA analysis. PCA component 1 accounted for 45.9% of variance and was mostly made up of shot location, while PCA component 2 accounted for 12.4% and evaluated a team's turnover tendencies and pace. Figure 2 visualizes our clusters over these two components. 3-point shooting to mid-range shooting ratios increase as you go left along the x-axis. Pace, turnover forcing, and fast break tendencies increase as you go up the y-axis.

6.2 Playstyle Clashes

While all playstyles have similar win-% (48%–52%) when they match-up against each other, the outcomes can be far from even. As shown in Figure 4, some playstyles have historically beaten other playstyles at higher rates. How-

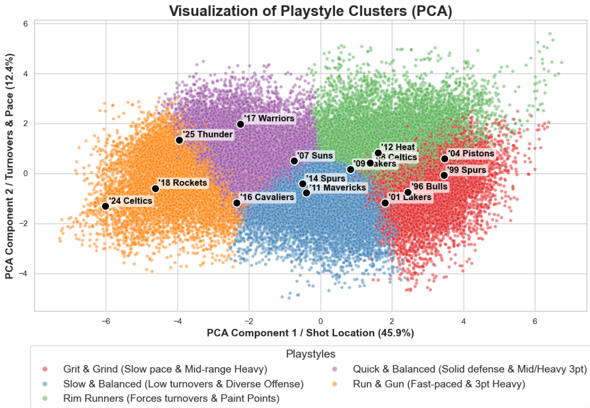

Figure 2: Playstyle mapped over Shot Location and Possession data

ever, it is important to account for some small sample sizes and potential confounding variables like team win %. Still, this seems promising for our theory that stylistic clashes have predictive power over wins.

7 Modeling

7.1 Creating Matchup Dataset

Before we began modeling, we had to once again refine our dataset. Crucially, we need to remove duplicate games from our data set. For each game, one row is dedicated to one team's statistics, meaning two rows explain the same game. We navigated this issue by splitting our dataset so that `team_A` was always the home

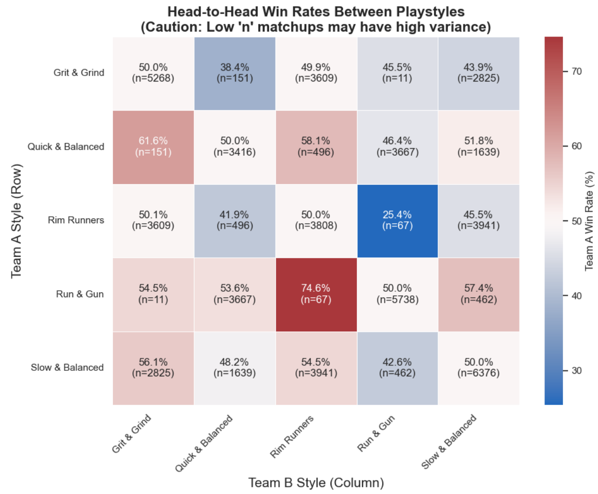

Figure 3: Win % between playstyles fluctuates on matchup.

team, and `team_B` was the away team. We then merged the home and away datasets together on `gameID`.

However, to eliminate the home team advantage that `team_A` would receive, we then shuffled the teams so that no home court advantage was detectable. A home column existed for each team which was 1 or 0 depending on if that team was at home, but these columns would be excluded for our advanced models. This would be crucial for some of our baseline modeling as well as our stylistic model.

Table 2: K-Means Cluster Assignments and Playstyle Archetypes

Cluster Label	**Key Aspects**
Quick & Balanced	Solid defensive metrics, moderate to heavy 3-point attempts, forces fast breaks.
Slow & Balanced	Low turnover percentage, diverse offensive mix, methodical half-court pace.
Grit & Grind	Very slow pace, heavy mid-range usage, low 3-point attempts, solid defense.
Rim Runners	Forces high opponent turnovers, moderate to heavy mid-range usage, strong paint point percentage, fast break generation.
Run & Gun	High pace, heavy 3-point attempt volume, very low mid-range, forces moderate turnovers.

7.2 Baseline Models

For our baselines, we chose 3 different simple models: home team always wins, higher win % always wins, and an offense focused model built with logistic regression. The first two were as simple as they sound, however the logistic regression did include a handful of feature columns. The features were the rolling offensive ratings and true shooting percentages for each team. This tested to see how effective purely offensive predictors were at evaluating game outcomes. These models were chosen because we decided that our model would be of no value if they could not outperform such simple evaluations.

7.3 Model Features

Three features are passed into our stylistic model. `win_percentage_rolling_10_A` and `win_percentage_rolling_10_B` are each team's 10-game rolling win rate going into the game, serving as the team quality control variable. And `style_matchup_encoded` is a label-encoded integer representing the specific cluster pairing (e.g. "2_vs_4" = Grit & Grind vs. Run & Gun), capturing the Rock-Paper-Scissors interaction directly. Under normal circumstances, using such limited variables would leave a model little room to improve, however our hypothesis would indicate that these 3 variables alone should hold a significant level of higher predictive power.

7.4 Training, Validation, and Test Split

For our model, we used a 3-way split on the data. 70% of data would go to training, and 15% each to validation and testing. Importantly, this split was chronological per season, so the training data comprised the first 70% of accessible games in each season, validation the next 15%, and testing the last 15%. This was necessary to prevent significant data leakage from chronological randomization. However, it does mean the model is vulnerable to significant mid/late season changes.

7.5 Hyperparameter Tuning

Rather than a grid search, we used Optuna with a Tree-structured Parzen Estimator (TPE) sampler to run 50 Bayesian optimization trials. The parameters we tuned are `max_depth` (3–8), `learning_rate` (0.05–0.3), `n_estimators` (300–1000), `subsample` (0.6–1.0), `colsample_bytree` (0.6–1.0), `min_child_weight` (1–10), `gamma` (0–1), `reg_alpha` (0–1), and `reg_lambda` (0.5–2.0). Each trial is evaluated by F1-score on the validation set with early stopping of 30 rounds. The best parameters are then used to train the final model.

7.6 Final Model Training

For our final model, we implement the tuned hyperparameters into XGBoost, since it was historically used in other sports modeling and was uniquely suited to our needs and features. The final XGBoost classifier is trained on `X_train` with `X_val` as the early stopping monitor. The model uses AUC as its eval metric during training and stops when validation AUC stops improving for 30 consecutive rounds. AUC, in this case, tells us how likely the model is to assign a higher winning probability to a winning `team_A` game over a losing `team_A` game. For our model, the test set is held out entirely until final evaluation.

8 Results

8.1 Baseline Performance

Before evaluating our XGBoost model, we established the performance for the three baselines to set a minimum bar for meaningful prediction. As shown in Table 4, Baseline 1 (home team always wins) achieved an accuracy of 0.5818 and a ROC-AUC of 0.5815, reflecting little more than the modest home court advantage present in the data. Baseline 2 (higher rolling win percentage wins) improved substantially to 0.6369 accuracy and 0.6301 ROC-AUC, confirming that recent team quality is the strongest single predictor of game outcomes. This has been a consistent finding across the broader NBA prediction literature. Baseline 3, a logistic regression model trained solely on rolling offensive rating and true shooting percentage, achieved 0.6082 accuracy but a comparatively strong ROC-AUC of 0.6542, suggesting that offensive efficiency carries meaningful probabilistic signal even without stylistic context.

8.2 XGBoost Model Performance

Our XGBoost model, trained on the style matchup encoding and both teams' rolling win percentages, achieved a test set accuracy of 0.6323, an F1-score of 0.6323, and a ROC-AUC of **0.6837**. The ROC-AUC of 0.6837 represents the strongest performance across all four models, surpassing Baseline 2 by 0.054 and Baseline 3 by 0.030. This is the most meaningful metric for our purposes, as it measures how reliably the model ranks a winning team's probability above a losing team's across all possible thresholds, independent of home/away imbalance.

On accuracy and F1, the results are more nuanced. Baseline 2 outperforms our model on both metrics (0.6369 and 0.6581 respectively), which at first glance appears to undermine our hypothesis. However, this is expected: Baseline 2 is a deterministic rule that exploits the strong signal in win percentage differentials, while our model must distribute its capacity across three features including the style matchup encoding. The fact that our model nearly matches Baseline 2 on accuracy while surpassing it substantially on ROC-AUC indicates that the style matchup feature is contributing genuine probabilistic signal, even if it does not consistently push hard binary predictions past Baseline 2's threshold rule.

8.3 Classification Balance: Home vs. Away

As shown in Table 3, the model predicts both away wins and home wins with nearly equal precision and recall (0.62–0.64 across both classes), indicating that the random team assignment in our matchup dataset successfully neutralized home court advantage. This symmetry is an important methodological validation: a model that systematically overpredicted home wins would suggest residual home court leakage rather than genuine stylistic signal.

8.4 Hypothesis Evaluation

Our first hypothesis, that K-Means clustering would recover meaningful, real-world basketball archetypes, is strongly supported. The five clusters align naturally with well-known NBA eras and team identities: Grit & Grind teams concentrate in the mid-2000s, Run & Gun teams dominate post-2016, and iconic franchises land in expected regions of the PCA space. The 2004 Pistons (defensive, slow-paced) and 2024 Celtics (fast, 3-point heavy) land at opposite ends of the PCA plot, providing strong face validity for the clustering solution.

Our second hypothesis, that playstyle interaction would add predictive power beyond simpler baselines, is only partially supported. The XGBoost model achieves the highest ROC-AUC of all models, suggesting that the style matchup encoding carries independent signal beyond win percentage alone. However, the model does not surpass Baseline 2 on accuracy or F1, which suggests that playstyle is best understood as a *secondary indicator* of game outcomes rather than a primary driver. Team quality, captured by rolling win percentage, remains the overwhelmingly dominant predictive force (Figure 4) a finding consistent with the broader sports analytics literature.

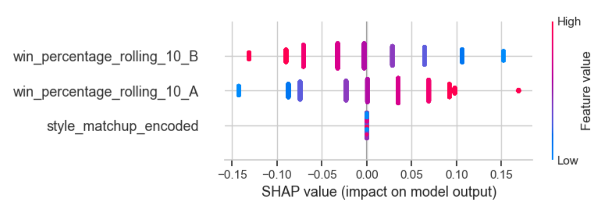

Figure 4: SHAP indicates the importance of features on the model's output.

Table 3: XGBoost Classification Report (Test Set)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Away Wins	0.62	0.64	0.63	2,161
Home Wins	0.64	0.62	0.63	2,229
Accuracy		0.63		4,390
Macro Avg	0.63	0.63	0.63	4,390
Weighted Avg	0.63	0.63	0.63	4,390

Table 4: Model Comparison on Test Set

Model	**Accuracy**	**F1-Score**	**ROC-AUC**
B1: Home Always Wins	0.5818	0.5934	0.5815
B2: Higher Win% Wins	0.6369	0.6581	0.6301
B3: LogReg (Offense Only)	0.6082	0.6183	0.6542
XGBoost (Style Matchup)	**0.6323**	**0.6323**	**0.6837**

9 Related Work

9.1 NBA Game Outcome Prediction

Predicting the outcome of NBA games is a well-studied problem in sports analytics. The most widely adopted approaches rely on Elo-based power ratings, where each team is assigned a continuously updated strength score and game outcomes are predicted from the difference in ratings. FiveThirtyEight's CARMELO and RAPTOR systems [1] are prominent examples, combining Elo ratings with player-level Box Plus/Minus estimates to generate win probabilities for individual matchups. Similarly, ESPN's Basketball Power Index (BPI) aggregates team and player efficiency metrics into a single predictive rating. While these systems are effective for season-long forecasting, they treat team strength as a single scalar value and make no attempt to model *how* teams play, only *how well* they perform on aggregate.

Machine learning approaches have expanded on rating systems by incorporating raw per-game and per-season statistics as features. Loeffelholz et al. [2] demonstrated that neural networks trained on season-average statistics could predict NBA game outcomes with approximately 74% accuracy. More recent work by Horvat and Job [3] applied ensemble methods including random forests and gradient boosted trees to similar feature sets, achieving comparable results. Thabtah et al. [4] explored logistic regression and naïve Bayes classifiers on boxscore statistics, finding that home court advantage and recent win percentage were consistently the strongest individual predictors across all model types. A common limitation across this body of work is that features are treated as independent predictors; the interaction between two teams' styles of play is never explicitly modeled.

9.2 Playstyle Analysis and Clustering in Sports

Clustering techniques have been applied to basketball to identify player archetypes. Murthy et al. applied K-Means to player tracking data to identify positional archetypes beyond the traditional five positions, finding that modern NBA players cluster into more granular roles such as "3&D wings" and "stretch bigs." Franks et al. [5] used spatial shot distribution data to characterize offensive styles at the player level. However, the application of clustering to *team-level* playstyle identification, and subsequently using those clusters as interaction features in a predictive model, has received far less attention in the literature. Most team-level analyses either manually define playstyle categories based on domain knowledge or use principal component analysis to reduce dimensionality without producing interpretable cluster labels.

9.3 Our Project's Niche

Our work sits at the intersection of these two streams. Like prior game prediction models, we use rolling efficiency statistics and win percentage as inputs. However, rather than treating each statistic as an independent feature, we first use K-Means clustering to derive discrete playstyle archetypes from the underlying statistical patterns, then encode the *interaction* between opposing archetypes as an explicit feature. This matchup-centric framing is the central novelty of our approach — it directly operationalizes the "rock-paper-scissors" hypothesis that certain playstyles hold structural advantages over others, independent of raw team quality. To our knowledge, no prior published work has modeled NBA game outcomes as a function of data-driven playstyle cluster interactions in this way.

Our finding that rolling win percentage remains the dominant predictor is consistent with the broader literature. However, the incremental improvement in ROC-AUC from our XGBoost model over the win-percentage-only baseline (B2) suggests that playstyle interaction captures genuine signal beyond team quality, a more granular conclusion than most binary outcome prediction papers report.

10 Conclusion

Our project found that stylistic clashes in the NBA actually bear very little predictive power over NBA game outcomes. While ROC-AUC value was at its highest in our stylistic model, it did not lead to increased accuracy in predicting which teams win or lose. This could mean that different playstyles don't have inherent advantages or disadvantages when evaluated amongst each other. Each playstyle seems to be viable if the team enacting it is of high enough quality. The "rock-paper-scissors" phenomena we theorized to be present in the NBA was not detected by our model. Instead, we reaffirmed the common finding that a team's recent win-% is a dominant driver in predicting NBA game outcomes.

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